

This problem is, however, mitigated by suitably including rain margins in the communication link design and also using C-band for heavy-rain-prone zones within the satellite footprint.

Mobile satellite communication

Satellite communication application was extended to mobile communication in the 1980s. Since the first experiments in satellite communication, those concerned with maritime and aeronautical communications had realised the potential benefits of satellite communication for their communities. Mobile satellite service in the beginning was considered primarily for maritime requirements. The arrival of cellular radio services evoked interest in land mobile satellite services as well. Due to the fact that mobile terminals are distributed over wide geographical areas, these services do not normally share the frequency bands used by fixed satellite links to avoid interference.

INMARSAT is an organisation that has provided mobile communication throughout the world in a big way. When INMARSAT was founded in 1979, it was mandatory to support services for maritime users only. During the 1980s, INMARSAT extended its field of interest to aeronautical and land mobile systems as well. The INMARSAT space segment is now deployed in the four-ocean region configuration, which was developed from the earlier three-ocean region arrangement designed to support maritime services. Fourth-generation satellites were deployed in 2005.

Since 1976, various INMARSAT systems for mobile communication have been operating, such as INMARSAT-A/B/C/M, INMARSAT-Aero and INMARSAT-D. INMARSAT-A provided circuit-switched telephony and telex services between PSTN and suitably equipped ships.

With the introduction of spot-beam capability provided by third-generation satellites, a derivative of



Inmarsat I-4 satellite

the INMARSAT-M, known as mini-M terminal, entered the service in the late 1996. This terminal, the land-mobile variant of which is about the size of a laptop computer, differs from full-M taking advantage of the higher gains of satellite antennae.

The INMARSAT M4 or global area network (GAN) uses land-portable terminals to provide on-demand voice, fax and 64kbps ISDN data services. GAN brings an ISDN environment to almost any remote location, and two GAN terminals can be multiplexed to achieve 128 kbps. The 3.1kHz audio mode allows analogue modem connections. A 2.4kbps circuit-switched data mode exists for backwards compatibility. The mobile packet data service (MPDS) is an always-on Internet/data service billed on the amount of data transferred instead of connection time. Typical applications of GAN include e-mail, Web surfing, transmitting large data files, real-time or store-and-forward video and secure communications. Most of the earth's land masses, except for the polar caps, are covered with this system.

The INMARSAT broadband global area network (BGAN) was launched in December 2005. It is the newest and most powerful INMARSAT service providing data speeds up to 492 kbps and simultaneous voice and data calling capability. BGAN offers on-demand guaranteed data rate 'stream-

ing IP' services up to 256 kbps, as well as 64kbps ISDN circuit-switched data for backwards compatibility with older GAN applications. INMARSAT terminals communicate in L-band frequencies.

INSAT MSS uses S-band for communication with mobile terminals. Under INSAT MSS, two services are operating; one for two-way voice and data communication using Type-A mobile terminal and the other for one-way reporting services using Type-C terminal. The reporting service is used for tracking vehicles and wagons on the move. Fixed links between the satellite and base stations of the MSS Type-A and Type-C services operate in the C-band.

The International Cospas Sarsat system is a satellite-aided search-and-rescue programme originally thought of for position location and rescue of ships that were mobile in the seas and in distress. Aeronautical and land components were also included in this system subsequently. Satellites in low-earth orbits and geostationary orbit are used for position determination in this system. The beacon designed to operate in VHF/UHF turns 'on' in distress situation and the signal transmitted is used in location determination.

Cospas and Sarsat satellites are in low-earth orbits and, as they fly past a distress beacon, observe the Doppler shift in frequency emanating from the